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Assessing the value of health information exchange organizations to hospital interoperability

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Abstract

Introduction: Various federal efforts have supported the development of local, regional, and state health information exchange organizations (HIOs), a type of network that promotes secure electronic exchange of patient health information among hospitals, healthcare providers, community-based organizations, and public health authorities. With the emergence of complementary and alternative modes to facilitate exchange, we sought to assess the value of HIOs to hospital interoperability.

Methods: We leveraged nationally representative survey data from a unique linked dataset of $N = 2200$ hospitals linked to HIOs to examine the relationship between hospitals' engagement in a range of interoperability activities and participation in an HIO that offers services to support those activities.

Results: We found that hospital participation in an HIO was significantly associated with greater engagement in clinical information exchange, public health reporting, and health-related social needs information exchange. However, among hospitals participating in an HIO, we did not find strong evidence of a relationship between specific HIO capabilities or services and hospital engagement in exchange.

Conclusion: Our findings indicate HIOs offer a useful avenue for exchange but are only one path through which exchange takes place as there are various alternatives available to support interoperability activities.

Key words: health information exchange; health information exchange organizations; interoperability; clinical information exchange; public health reporting; health-related social needs; national networks.

Introduction

Support for local, regional, and state health information exchange organizations (HIOs), a type of network that enables health information exchange across different entities, was an important component of the Health Information Technology for Economic and Clinical Health (HITECH) Act of 2009, highlighted by the State Health Information Exchange Cooperative Agreement Grants totaling over $500 million.[1-3] As of 2023, approximately 135 HIOs existed with very different capabilities to support exchange.[4] Most hospitals report using multiple methods for exchange in addition to a HIO, including prominent national health information exchange networks that connect healthcare providers, HIOs, public health authorities (PHAs), and payers (eg, CommonWell Health Alliance, e-Health Exchange, Carequality) and electronic health record (EHR) vendor networks (eg, Epic's Care Everywhere), to support a range of interoperability-related activities.[5,6] Alternatives to HIOs also exist to support exchange in specific domains (eg, social care referrals or facilitating integration of apps with the EHR).[7,8]

Within this context, we sought to assess the degree to which HIOs impact hospitals' engagement in interoperability. Understanding how impactful HIOs are to hospital interoperability can help inform state and federal strategies, such as the Trusted Exchange Framework and Common Agreement™ (TEFCA™), a nationwide interoperability framework, authorized by Congress in the 21st Century Cures Act and launched in late 2023, designed to connect "health information networks," inclusive of HIOs and other networks, through a common technical and policy foundation.[9] TEFCA is intended to promote the secure exchange of standardized data across health information networks and thus is likely to accelerate existing trends towards connectivity across networks and bring opportunities and challenges for HIOs.[4] Information on how HIOs' plans to participate in TEFCA impacted hospitals' plans could provide useful information on the likely evolution of nationwide information exchange through TEFCA and the options available to hospitals to participate in nationwide exchange.

To assess the value of HIOs to hospital interoperability as TEFCA went live, we leveraged nationally representative survey data from hospitals and state, regional, or local HIOs to examine whether hospital participation in an HIO and hospital participation in an HIO that offered specific services were associated with greater engagement in clinical information exchange, reporting to PHAs, and exchange of data on individuals' health-related social needs (HRSN), such as reliable transportation, housing, and access to healthy food. We also examined whether HIOs' plans to participate in TEFCA influenced hospitals' awareness or plans to participate in TEFCA. Specifically, we investigated whether:

1. Clinical information exchange: Hospitals that participate in an HIO that participates in a national network and

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routinely makes data available in a structured format were more likely to report often engaging in clinical information exchange (the ability to find, send, receive, integrate information from outside sources).
2. Reporting to PHAs: Hospitals that participate in an HIO that supports public health reporting were more likely to electronically report information to PHAs.
3. HRSN information exchange: Hospitals that participate in an HIO that makes HRSN data available and have capabilities or services to support HRSN information exchange were more likely to report electronically receiving or using HRSN data to inform patient care.
4. TEFCA participation: Hospitals that participate in an HIO that plans to participate in TEFCA were more likely to be aware of TEFCA or plan to participate, and whether that varied by hospital participation in national networks.

Data and methods

Data

This study leveraged a unique linked dataset of U.S. hospitals and HIOs. Hospital data came from the 2023 American Hospital Association (AHA) Information Technology (IT) Supplement, a nationally representative survey of U.S. hospitals fielded March to August 2023. This study was limited to non-federal acute care hospitals ($N = 2{,}547$, response rate = 58%), which have historically been the target of incentive programs aimed at promoting interoperability. HIO data came from the 2023 HIO Survey, a near-census of HIOs led by the Office of the Assistant Secretary for Technology Policy in collaboration with Civitas Networks for Health and the University of California San Francisco. The survey was fielded January to July 2023. The response rate for eligible HIOs was 86% ($N = 77$); responding HIOs operated in 47 states and the District of Columbia.

HIO name was used to merge responses from the HIO survey into the AHA IT Supplement. Of the 2547 non-federal acute care hospital respondents to the IT Supplement, 490 did not participate in an HIO and 1710 had at least 1 match in the HIO survey. The final analytic sample therefore included 2200 non-federal acute care hospitals. 347 hospitals that responded to the AHA IT Supplement were excluded because they reported participating in an HIO but did not name that HIO or named an HIO that could not be matched to the HIO survey. For all analyses, we reported on the share of hospitals connected to *at least 1* HIO. Details on the data linkage and study sample are available in Appendix A.

Measures

Outcome measures came from the AHA IT Supplement. Hospitals' engagement in clinical information exchange was measured by the hospital reporting that they often or routinely find, send, receive, or integrate patient health information from sources outside their hospital or health system, or engage in all 4 domains. Hospitals' engagement in public health reporting was measured by hospital-reported use of electronic methods to submit data for syndromic surveillance, immunization registry, case reporting, laboratory (lab) result reporting, public health registry, and clinical data registry reporting. Hospitals' engagement in HRSN information exchange was measured by hospital-reported receipt of HRSN data and subsequent use of data received for internal purposes—to make referrals to social service organizations, inform clinical decision-making, and discharge planning—as well as for secondary uses—to conduct population health analytics and inform community needs assessment or other equity initiatives. We also captured hospitals' awareness of and plans to participate in TEFCA.

Our main independent variable of interest came from the AHA IT Supplement and indicated hospitals that participated in at least 1 HIO. We examined the relationship between HIO participation and all outcomes. Other key independent variables of interest came from the HIO survey and were mapped to specific groups of outcomes. For each clinical information exchange outcome, independent variables included whether a connected HIO participated in a national network or made patient health information available in a structured format (ie, structured care summaries and data consistent with the United States Core Data for Interoperability (USCDI) v1/v2 and using the Health Level Seven (HL7®) Fast Healthcare Interoperability Resources® (FHIR®) standard). For each public health reporting outcome, independent variables included whether a connected HIO was connected to a state or local PHA and offered public health reporting services to support that specific reporting type (eg, HIO services to support electronic case reporting were mapped to the electronic case reporting outcome). For each HRSN information exchange outcome, independent variables included whether a connected HIO made HRSN data available as part of a clinical document or as a structured data element, used a social service referral platform (such as Unite Us or Aunt Bertha), offered closed-loop referrals tracking that could provide hospitals with updated information on referrals that originated in the hospital and progressed to a community-based organization or social service setting, or identified gaps in care. We also included a variable indicating whether a connected HIO planned to participate in TEFCA as a Qualified Health Information Network® (QHIN™), participant, or sub-participant, which may influence hospitals' awareness of or plans to participate in TEFCA. Appendix Table B1 shows how each independent variable maps to each measure category. Survey questions used to create key measures are available in Appendix C.

Covariates came from the AHA Annual Survey and included hospital size, ownership, teaching status, system affiliation, location (urban vs rural), critical access designation, and participation in value-based care (participation in an accountable care organization, patient-centered medical home program, or pay for performance arrangement). We also controlled for hospital participation in at least 1 major national network (CommonWell Health Alliance, e-Health Exchange, or Carequality) or EHR vendor-based network.

Analysis

First, we described the share of hospitals that participated in an HIO, and the share connected to an HIO that had capabilities or services to facilitate clinical information exchange, public health reporting, HRSN information exchange, or participation in TEFCA. We then examined the relationship between these outcomes of interest by HIO participation and the availability of specific HIO services to support that activity using separate weighted linear probability models, adjusted for hospital characteristics and participation in a national network or EHR vendor-based network. All results were weighted to account for non-response and to generate national estimates. A logistic regression model was used to predict the propensity of survey response as a function of hospital characteristics, including size, ownership, teaching status, system affiliation,

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and availability of a cardiac intensive care unit, location, and region. Hospital-level weights were derived by the inverse of the predicted propensity. For all models, P values < 0.05 were considered statistically significant. Data analyses were performed between August 2024 and February 2025 using Stata/SE, version 15.1 (StataCorp).

Results

Availability of HIO services to facilitate information exchange

In 2023, 75% of hospitals participated in an HIO (Appendix Table B1). More than half (59%) were connected to an HIO that participated in a national network and routinely made available care summaries in a structured format (54%). Fewer hospitals were connected to an HIO that routinely exchanged information using the FHIR standard (17%) and routinely exchanged information consistent with USCDI v1 or v2 (36%). About three-quarters of hospitals participated in an HIO that was connected to at least 1 state or local PHA (74%). However, only some of these hospitals were connected to an HIO that offered the following reporting services to PHAs: immunization registry (54%), syndromic surveillance (53%), electronic lab reporting (50%), electronic case reporting (30%), public health registry (40%), and clinical data registry reporting (45%). While half of hospitals (48%) participated in an HIO that made HRSN data available, fewer participated in an HIO that used a social service referral platform (16%), offered closed-loop referrals tracking (32%), or identified gaps in care (32%). About half of hospitals (53%) participated in an HIO that planned to participate in TEFCA.

Clinical information exchange

Hospitals that participated in an HIO were more likely to report engagement in each of the 4 domains of clinical information exchange individually, but not collectively, compared with non-participants (eg, 79% vs 67% for finding information) (Figure 1, Appendix Table B3). There were no significant differences in hospitals' engagement in clinical information exchange by participation in an HIO that participated in a national network.

Evidence of a relationship between hospital engagement in interoperable exchange and participation in an HIO that routinely made information available in a structured or standardized format was mixed. Hospitals connected to an HIO that sent structured care summaries were less likely to report routinely receiving and integrating information compared with hospitals that participated in an HIO that did not make structured care summaries available (78% vs 82% and 60% vs 66%, respectively). Conversely, hospitals connected to an HIO that routinely sent data consistent with USCDI v1 or v2 were more likely to report routinely integrating information (66% vs 57%) compared with hospitals that participated in an HIO that did not make data available consistent with USCDI. Hospitals connected to an HIO that routinely shared data using the FHIR standard were also more likely to report integrating and being engaged in all 4 domains of clinical information exchange compared with hospitals connected to an HIO that did not make data available using the FHIR standard (68% vs 60% and 62% vs 48%, respectively). There were no significant differences in rates of hospitals finding or sending information by participation in an HIO that made data available in a structured format.

Public health reporting

Hospitals participating in an HIO were more likely to report engagement in each of the 6 types of electronic public health reporting compared with non-participants (eg, 97% vs 91% for immunization registry reporting) (Figure 2, Appendix Table B5). Hospitals connected to an HIO that offered services to support public health reporting were also more likely to be

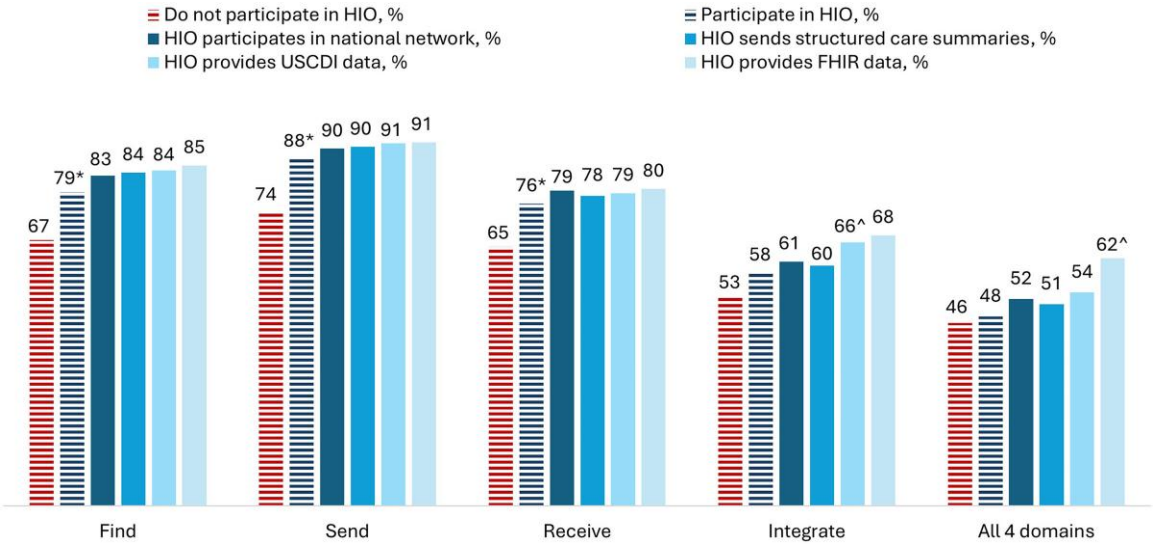

Figure 1. Hospital engagement in clinical information exchange, by participation in an HIO with capabilities or services to support clinical information exchange. Source: 2023 AHA IT Supplement and 2023 HIO Survey. All percentages are weighted percentages adjusted for hospital characteristics and participation in a national network or EHR vendor-based network. *Indicates significantly higher rates of hospitals' engagement in clinical information exchange among HIO participants relative to non-participants. ^Indicates significantly higher rates of hospitals' engagement in clinical information exchange among participants whose HIO offers a specific service or capability to support exchange relative to those whose HIO does not offer that service or capability. Statistical significance was assessed at the 5% level ($P < 0.05$).

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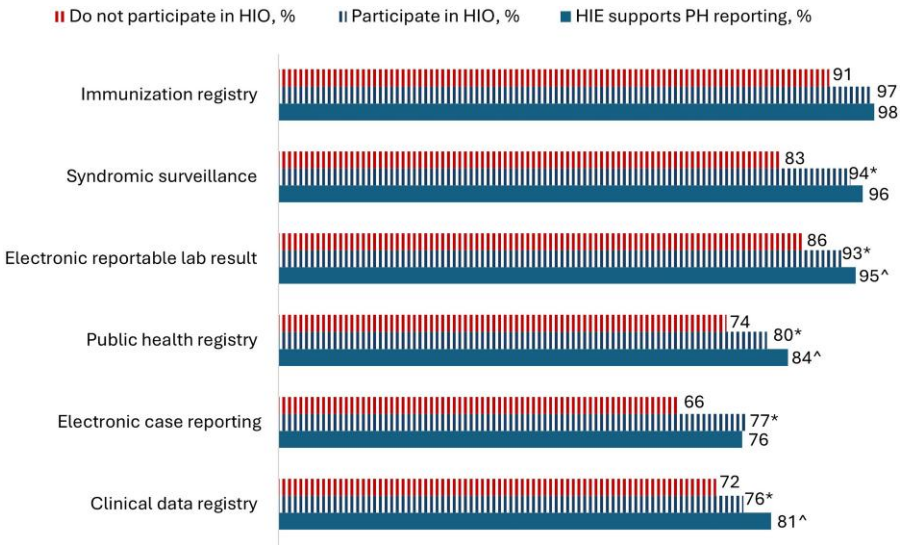

Figure 2. Hospital engagement in electronic public health reporting, by participation in an HIO with capabilities or services to support public health reporting. Source: 2023 AHA IT Supplement and 2023 HIO Survey. All percentages are weighted percentages adjusted for hospital characteristics and participation in a national network or vendor-based network. *Indicates significantly higher rates of hospitals' engagement in electronic public health reporting among HIO participants relative to non-participants. ^Indicates significantly higher rates of hospitals' engagement in electronic public health reporting among participants whose HIO offers services to support that reporting type relative to those whose HIO does not offer that service. Statistical significance was assessed at the 5% level ($P < 0.05$).

engaged in electronic public health reporting, but only for electronic reportable lab result (95% vs 90%), public health registry (84% vs 79%) and clinical data registry (81% vs 74%) reporting.

HRSN information exchange

Hospitals participating in an HIO were more likely to report receiving data from outside sources compared with non-participants (60% vs 51%). HIO participants were also more likely to use data received from outside sources for different purposes compared with non-participants (eg, 45% vs 35% for making referrals, 37% vs 30% for population health analytics) (Figure 3, Appendix Table B7).

However, evidence of a relationship between hospitals' receipt and use of HRSN data and participation in an HIO that offered services to support HRSN information exchange was mixed. Hospitals that participated in an HIO that made HRSN data available were more likely to report using data to make referrals (50% vs 43%) and inform community needs assessments (37% vs 32%), yet less likely to report using data for clinical decision-making (42% vs 52%), discharge planning (39% vs 53%), and population health analytics (38% vs 43%) (Appendix Table B7). Hospitals connected to an HIO that used a social service referral platform were more likely to report using data to inform discharge planning (49% vs 43%) yet less likely to report using data to make referrals (43% vs 49%). Compared with hospitals connected to an HIO that did not offer these services, hospitals connected to an HIO that identified gaps in care were more likely to report using data to inform discharge planning (50% vs 39%).

Planned participation in TEFCA

Hospitals participating in an HIO were more likely to report being aware of TEFCA and have plans to participate compared with non-participants (79% vs 59% and 65% vs 45%, respectively) (Figure 4, Appendix Table B9). Hospitals connected to an HIO that planned to participate in TEFCA were also more likely to be aware of TEFCA and plan to participate compared with those connected to an HIO without plans to participate in TEFCA (84% vs 78% and 70% vs 64%, respectively).

Rates of TEFCA awareness and plans to participate were higher among HIO participants relative to non-participants, regardless of their participation in a broader national network. For example, among hospitals participating in e-Health exchange, HIO participants were more likely to be aware of and plan to participate in TEFCA compared with non-participants (97% vs 89% and 93% vs 80%, respectively) (Appendix Table B9). Findings were similar for hospitals participating in other national networks and EHR vendor-based networks.

Discussion

In 2023, three-quarters of non-federal acute care hospitals in the U.S. were connected to at least 1 state, regional, or local HIO. Our findings suggest that hospital participation in an HIO was associated with greater engagement in clinical information exchange, public health reporting, and receipt and use of HRSN data. Participation in an HIO was also associated with a greater likelihood of hospitals having plans to participate in TEFCA, regardless of their participation in a broader national network. However, our findings do not provide strong evidence of a relationship between specific services offered by HIOs to support health information exchange and hospitals' engagement in exchange. These findings suggest that HIOs offer value to hospital interoperability activities, but that hospitals are not dependent on HIOs to support interoperable exchange.

Hospital connectivity to HIOs was associated with 6–12 percentage-point greater rates of routine engagement in clinical information exchange, which suggests HIO participation may help facilitate some hospitals' ability to receive and share data with other providers. However, we observed no difference in

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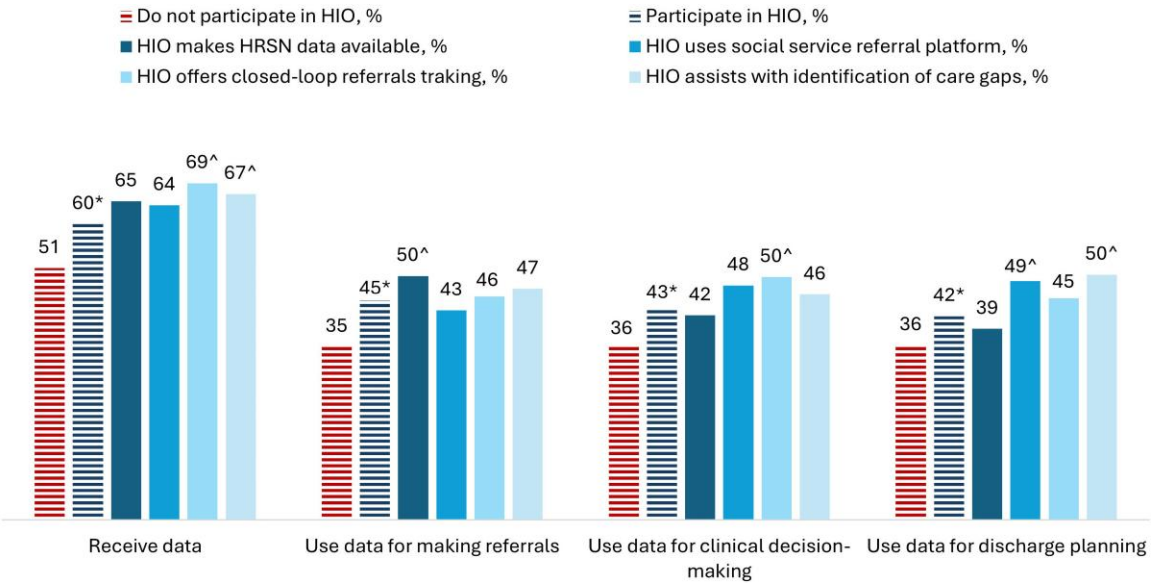

Figure 3. Hospital engagement in HRSN information exchange, by participation in an HIO with capabilities or services to support HRSN information exchange. Source: 2023 AHA IT Supplement and 2023 HIO Survey. All percentages are weighted percentages adjusted for hospital characteristics and participation in a national network or vendor-based network. This figure illustrates the relationship between HIO participation and hospitals' receipt and use of data for internal purposes (making referrals, clinical-decision making, discharge planning); estimates of the relationship between HIO participation and hospitals' use of data for secondary purposes (informing community needs assessment and population health analytics) are reported in Appendix Table B7. *Indicates significantly higher rates of hospitals' engagement in HRSN information exchange among HIO participants relative to non-participants. ^Indicates significantly higher rates of hospitals' engagement in HRSN information exchange among participants whose HIO offers a specific service or capability to support exchange relative to those whose HIO does not offer that service or capability. Statistical significance was assessed at the 5% level ($P < 0.05$).

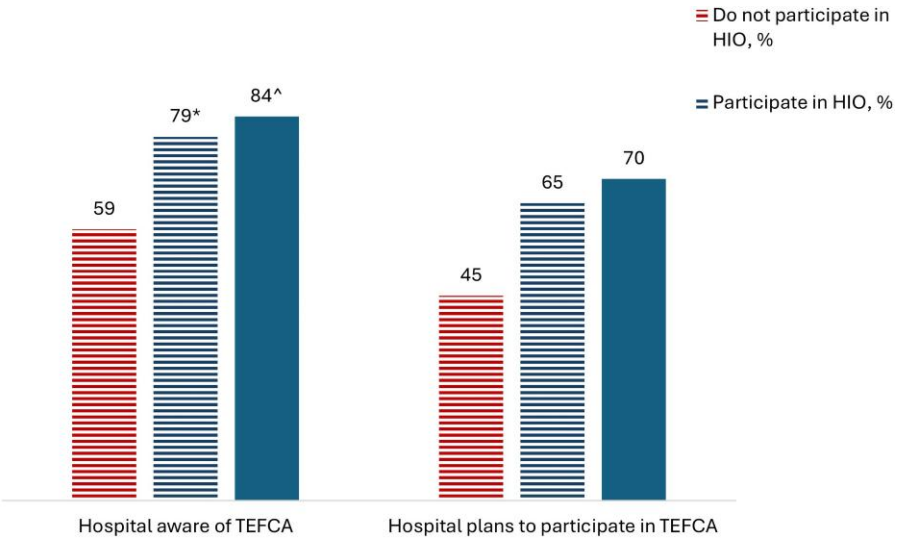

Figure 4. Hospitals' awareness of and planned participation in TEFCA, by participation in an HIO that plans to participate in TEFCA. Source: 2023 AHA IT Supplement and 2023 HIO Survey. All percentages are weighted percentages adjusted for hospital characteristics and participation in a national network or vendor-based network. *Indicates significantly higher rates of hospitals' awareness or plans to participate in TEFCA among HIO participants relative to non-participants. ^Indicates significantly higher rates of hospitals' awareness or plans to participate in TEFCA among participants whose HIO plans to participate in TEFCA relative to those whose HIO does not have plans to participate in TEFCA. Statistical significance was assessed at the 5% level ($P < 0.05$).

engagement in exchange in hospitals that participated in HIOs that did or did not support structured care summaries—a core capability related to exchange. One interpretation of this dynamic, and similar ones we observe in other domains, is that hospitals that participate in an HIO are also otherwise more likely to engage in exchange through other means, inflating the association between HIO participation broadly and engagement in clinical information exchange. In contrast, we did observe a difference between hospitals participating in HIOs with advanced capabilities to support interoperable exchange—through their support for USCDI and FHIR-based exchange—who were more likely to be engaged in clinical care exchange than hospitals participating in HIOs without those capabilities.

Beyond facilitating clinical information exchange, HIO services have increasingly evolved to facilitate exchange

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between healthcare and public health. During the COVID-19 pandemic, there was significant federal investment in enhancing HIO services to support electronic reporting to PHAs, much of which was focused on improving the timeliness, accuracy, and completeness of lab and case reporting.[10] In our study, HIO participation was associated with higher rates of engagement in all types of public health reporting. Further, hospitals connected to an HIO that supported electronic lab reporting had significantly higher rates of reporting compared with hospitals connected to an HIO that did not offer this service. We did not see similar results for electronic case reporting, which has faced more substantial challenges to adoption despite rapid growth in use during the COVID-19 pandemic.[11] These findings suggest HIO services are utilized for lab reports more than case reports, likely due to electronic lab reporting being widely used by both hospitals and PHAs.[12] The availability of HIO services to support public health reporting made no difference for the most common reporting types (immunization and syndromic surveillance), which suggests there may be preferred alternatives for submitting these data types. However, our findings indicate that HIOs may facilitate reporting for less common reporting types (eg, clinical data registry reporting), perhaps by alleviating reporting challenges, such as the technical complexity of interfaces, transmission, or submission.[13]

Evidence of a relationship between hospitals' receipt and use of HRSN data and the availability of specific HIO services to support HRSN information exchange was even more mixed. While HIO participants were consistently more likely to report receiving and using HRSN data from outside sources, there were no consistent signals to suggest services offered by HIOs—including making HRSN data available to participants—facilitated hospitals' engagement in HRSN information exchange. It is possible that hospitals are utilizing these services directly (not mediated by an HIO) or have established other processes for receiving HRSN data to facilitate different uses, such that these services offered by HIOs do not drive differences in receipt or use. It is also possible that organization-specific culture and ethics related to the use of HRSN are stronger drivers on the decision to use HRSN data from outside the organization than data availability through HIOs, resulting in these limited correlations.

Finally, we found that hospitals participating in an HIO that planned to participate in TEFCA were more likely to be aware of TEFCA and have their own plans to participate, regardless of hospitals' participation in a national network, which may also affect hospitals' awareness or plans to participate since some of these are QHINs. While TEFCA provides opportunities for all hospitals to participate in streamlined exchange regardless of participation in an HIO, our findings suggest HIO participation may uniquely contribute to hospitals' readiness to participate in TEFCA, possibly because HIOs could facilitate hospital participation in TEFCA as a participant or subparticipant to the HIO. However, the overall magnitude of this association was relatively small with 64% of hospitals that participated in an HIO that did *not* plan to participate in TEFCA nevertheless planning to participate, indicating the salience of other options. Because TEFCA participation is voluntary, it will be important to monitor whether HIOs are effective at supporting participation by hospitals and other healthcare organizations, what types of healthcare organizations choose not to participate in TEFCA even when provided the option by their HIO, and which healthcare organizations lack a viable path to participate.

Limitations

The data used for this study represent self-reports of hospitals' engagement in health information exchange and HIO capabilities or services to support exchange at a single point in time. We were not able to verify actual HIO capabilities or availability of services at the time in which hospitals would have used them. Further, availability of HIO services does not reflect actual use by hospitals. It's possible that hospitals did not use or were not aware of specific services made available to them through the HIO(s) in which they participated. While all estimates were adjusted for observable hospital characteristics and participation in a national network or EHR vendor-based network, it is possible that hospitals with the resources and technical capabilities to exchange were more likely to self-select into HIO participation.

Our findings demonstrate an association between hospital participation in HIOs and engagement in exchange; we do not establish a causal relationship between HIO participation and hospital interoperability. It is also possible the value HIOs add to hospital interoperability is not fully captured by higher rates of exchange, but in better data quality (ie, improvements in the completeness of records). Finally, conclusions from this survey indicate HIOs' value to hospital interoperability; HIOs may offer more or less value to other healthcare providers, community-based organizations, or other entities.

Conclusion

Using a unique linked dataset of hospitals and HIOs, this study provides evidence to suggest HIOs may help facilitate hospitals' ability to receive and share data with other providers and PHAs. Our findings indicate that hospital participation in HIOs is common and was significantly associated with greater engagement in clinical information exchange, public health reporting, HRSN information exchange, as well as plans to participate in TEFCA. However, in many cases, hospitals participating in HIOs that offered specific supportive services did not participate in exchange at greater rates. These dynamics indicate that participation in a HIO with specific capabilities may have limited impact because hospitals are likely to engage in interoperability through alternative networks or by leveraging other technologies. Based on our findings, HIOs appear to offer a useful avenue for exchange and reporting activities but are only one path through which this exchange takes place.

Supplementary material

Supplementary material is available at *Health Affairs Scholar* online.

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Conflicts of interest

The authors do not have any conflicts of interest to disclose. Please see ICMJE form(s) for author conflicts of interest. These have been provided as supplementary materials.

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